Random walks as a non-uniform sampling strategy in R^D for optimal experimental design conditions

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Project Objective

The development of a new, non-uniform sampling (NUS) strategy in R^D. The early stage of research included the exploration of various NUS strategies currently in development, the problems and fields they were developed to address, and the costs and benefits of such strategies. Then, I explored the theoretical and practical implementation of such strategies. Finally, I decided that a random walk method was an effective strategy for this problem. My strategy applied to any higher dimension of R^D and considered all of a D-cube region of R^D.

Introduction

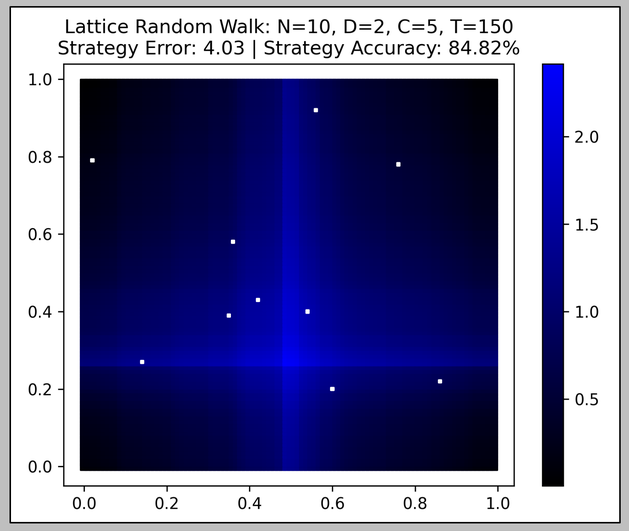
Scientific research involves vast amounts of data samples. Sample collection involves many costs, including technology usage, material expenditure, and data collection time. As modern research grows computationally and financially more expensive, minimizing input costs is vital to conducting quality research while creating experimental designs within budget. In the field of optimization, NUS strategies offer a cost-efficient alternative to finding the best conditions under which to make decisions. In practice, NUS strategies can be used to find the best experimental design conditions that explore all variables associated with a high dimensional probability field. NUS strategies reduce the cost of research by finding a non-uniform distribution of points for which to develop varied and practical experimental designs that yield productive research outcomes.

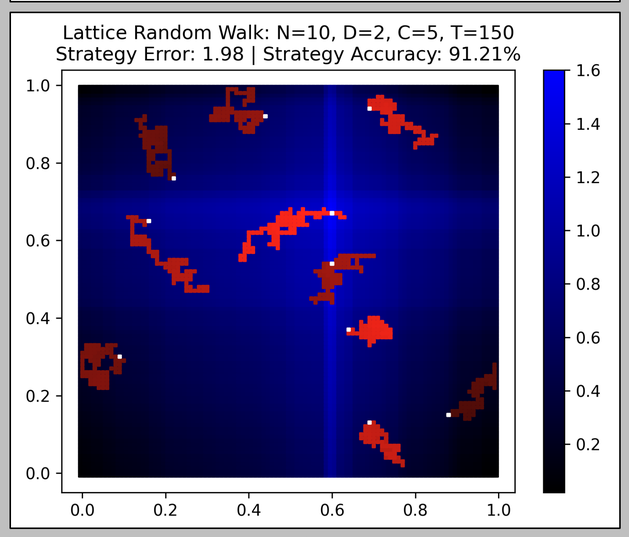
References

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Random Walk Theory

There are many different types of random walks. For this project, a lattice random walk model was used to move samples through a D-cube region of R^D. Two directions were defined along each dimension, resulting in 2·D possible directions. This lattice random walk had a fixed step size of 1 unit, with a random direction chosen after each step. Each random direction had an equal probability of $\frac{1}{2 \cdot D}$. This way, each random step was independent and identically distributed. Under these conditions, by the Central Limit Theorem for random walks, the possible distances from the origin of a random walk would converge to a normal distribution as the number of steps increased. This led to semi-predictable long-term behavior and the expected distance a random walk would travel from an origin, known as the root mean square, or $\sqrt{T}$ for T steps.





Control Set	
Constant Values	
Step size (S)	1
Region length per axis (L)	1
Subdivisions per axis (A)	100
Mean range (Rm)	[0, 1]
Standard deviation range (Rd)	[0.1, 1]
Trials (X)	100
Variable Lower Bounds	
Samples (N)	10
Dimensions (D)	3
Clusters per axis (C)	3
Steps (T)	1

Independent Experiments			
Treatment	N	D	C
1	20	4	4
2	30	5	5
3	40	6	6
4	50	7	7
5	60	8	8
6	70	9	9
7	80	10	10
8	90	11	11
9	100	12	12

Methods

The probability field was generated over the lattice region as the average of D normal curves along each axis. Normal curves were generated with random mean and standard deviation values. N samples were generated with origin placements according to a Latin Hypercube Sampling distribution generated by dividing the lattice region into N subdivisions along each axis. Random walks operated for T steps. For each random walk, the sample with the highest probability value was selected as the best sample B. These B samples were the optimal conditions for experimental design.

Strategy accuracy was measured by dividing each axis into C parts, resulting in C^D clusters. Each cluster selected a random probability P from its region. For all clusters that contained B samples, the total error was

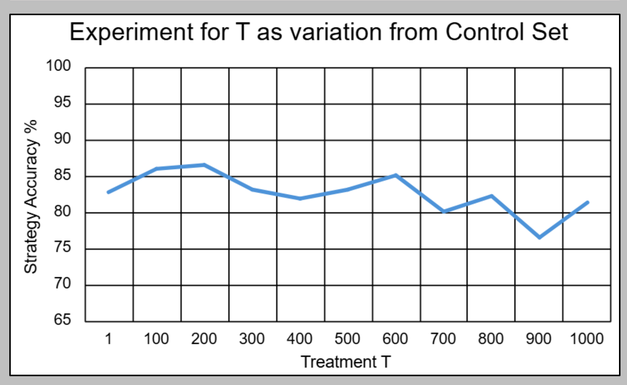
$$error_{cluster} = \sum |B_{cluster} - N \cdot P_{cluster} \cdot \frac{1}{\sum P}| \ .$$

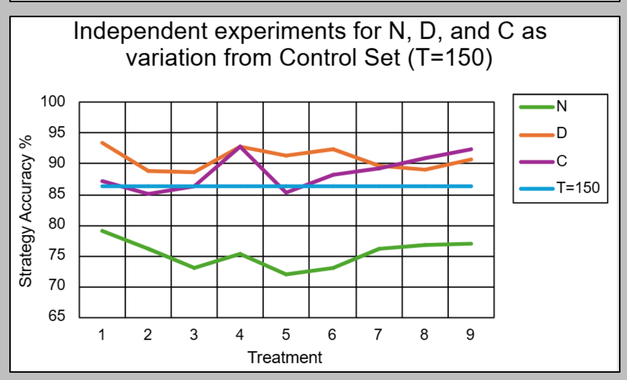
The worst possible error was

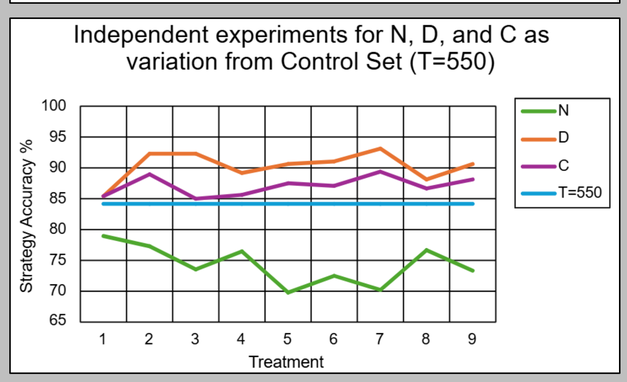
$$error_{worst} = 2 \cdot N \cdot (1 - P_{highest} \cdot \frac{1}{\sum P}) \ .$$

The strategy accuracy was

$$accuracy = 1 - error_{cluster} \cdot (error_{worst})^{(-1)} \ .$$







Results

Each strategy accuracy data point was the average of 500 random walk trials per treatment. Each trial had constant values and variable lower bounds set according to the Control Set. Experiment for T revealed T=100 to T=200 and T=500 to T=600 maintained nearly 85% or more strategy accuracy. Across T=150 and T=550, independent experiments for N, D, and C revealed consistent trends of strategy accuracy behavior per variable per treatment.

Discussion

These results suggest between T=100 and T=200 are some of the best conditions under which random walks could consistently yield high strategy accuracy percentages, with root mean squares between 10 and 15. The lattice region had 100 subdivisions per axis, which suggests that root mean squares between 10% and 15% of a region length could be consistently more productive for random walks in R^D to find optimal experimental design conditions.

Future studies could investigate strategy accuracy as treatments for N increase. A major program optimization to be made pertains to the cluster generation method per trial. If complete, experiments with high T, N, D, and C values could be computed in reasonable time. Other strategies including gradient bias and subsampling, Voronoi diagrams, and Markov Chain Monte Carlo methods were considered as potential NUS strategies. Overall, random walks were the most accessible in my academic journey so far.

Acknowledgements

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